Kieu Dang

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RESEARCH INTERESTS

Trustworthy AI, with a focus on privacy, security, and robustness in LLMs. My work explores adversarial robustness, differential privacy, and watermarking to ensure reliable and ethical deployment of LLMs in real-world applications.

EDUCATION

State University of New York at Albany (SUNY Albany), NY, USA. Aug 2024 – Present
PhD Candidate in Information Science. GPA: 4.00/4.00
Research topics: Trustworthy AI through the Lens of Privacy and Security.

Northeastern University, MA, USA. Aug 2021 – July 2023
Master in Analytics, Applied Machine Intelligence. GPA: 4.00/4.00
Thesis: Improving Safety through the Integration of Multi-sensor Fusion and Deep learning-based Object Detection.

PUBLICATIONS

- **Kieu Dang** Huy Phan, Ojaswi Dulal, Aiham AL Shukairi, Abby Shine, Chase Garner, Phung Lai. LLM Watermarking as Big Data Provenance: A Deployment-Oriented Systematization - In *Proceedings of the 13th IEEE Special Session on Privacy and Security of Big Data, IEEE BigData* **(IEEE PSBD 2026)**.
- **Kieu Dang**, Phung Lai, NhatHai Phan, Yelong Shen, Ruoming Jin, Abdallah Khreishah. δ-Steal: LLM Stealing Attack with Local Differential Privacy. In *Proceedings of Asian Conference on Machine Learning* **(ACML 2025)**.
- Dylan Tarace, Phung Lai, **Kieu Dang**, Unal Tatar. AI-Powered Assessment of Wazuh for Obfuscated Threat Detection. In *Proceedings of the IEEE Systems and Information Engineering Design Symposium* **(SIEDS 2025)**.
- **Kieu Dang**, Phung Lai. Navigating Trustworthiness in LLMs: An Examination of Privacy, Security, and Robustness. In *Proceedings of Computational Data and Social Networks* **(CSoNet 2024)**.

Forthcoming

- **Kieu Dang**, Phung Lai, Ching-Yun Ko, Pin-Yu Chen. *Adaptive Co-Serving LLM Watermarking on Modern Inference Engines* - **Under review**.
- **Kieu Dang**, Phung Lai, NhatHai Phan, Yelong Shen, Ruoming Jin. *Robust LLM Watermarking with Minimal Semantic Distortion for IP Protection* - **Under review**.
- **Kieu Dang**, Phung Lai, NhatHai Phan, Yelong Shen, Ruoming Jin, Abdallah Khreishah, My Thai. *SoK: Are Watermarks in LLMs Ready for Deployment?* - **Under review**.

PATENTS

- **Kieu Dang**, Phung Lai, NhatHai Phan. **SWIFT watermark: Adaptive Co-Serving LLM Watermarking on Modern Inference Engines**. Provisional US patent filed in October 2026.
- **Kieu Dang**, Phung Lai, NhatHai Phan. **SafeSeal watermark: Robust LLM Watermarking with Minimal Semantic Distortion for IP Protection**. Non-provisional US patent accepted on October 8, 2025.

RESEARCH EXPERIENCE

IBM Research, USA. Jan 2026 – Present
Research Collaborator

- Collaborate with IBM Research on trustworthy generative AI, developing scalable LLM watermarking methods integrated into inference engines for efficient provenance tracking.
- Investigate provenance and attribution mechanisms for agentic AI workflows and multimodal AI systems.

Responsible AI Lab - SUNY Albany, NY, USA. Jan 2024 – Present
Research Assistant

- Cooperate with Microsoft, New Jersey Institute of Technology, University of Florida, Kent State University, Qatar Computing Research Institute to prove theories, implement, and conduct experiments for Trustworthiness in AI.
- Design and evaluate strategies that strengthen LLMs against model stealing, incorporating watermarking and privacy-preserving techniques.

TEACHING & STUDENT MENTORSHIP EXPERIENCE

SUNY Albany, NY, USA Jan 2025 – Present
Teaching Assistant and Student Mentor

- Teaching Assistant for *CYBR-422: Trustworthy AI*: conducting labs, grading, and providing student support.
- Mentor for undergraduate interns on multiple research projects, including *LLM watermarking, human-based evaluation system, and public leaderboard*, *Transforming LLM Alignment: Automating Reference Data Generation through Explainable AI*, and *Powered Assessment of Wazuh for Obfuscated Threat Detection.*
- Served as Chancellor's Summer Research Excellence Mentor, supervising six undergraduate interns across AI security projects: on LLM watermarking, human-based evaluation system, and public leaderboard.

Northeastern University, ON, Canada. Aug 2021 – Dec 2021
Teaching Assistant: *ALY-6010: Introduction to Statistics and Probability:* Provided tutoring support and evaluated student performance through discussion grading.

PROFESSIONAL EXPERIENCE

A Medium Corporation, CA, USA. Nov 2022 – Present
Machine Learning Content Specialist (Freelancer – Partner Program)
Translate complex machine learning topics, especially in NLP, into accessible and insightful content for a diverse audience.

Hitachi Vantara Corporation, CA, USA. Oct 2023 – May 2024
Senior Data Scientist (Remote Full Time)

- Spearheaded NLP initiatives to enhance financial document analysis, improving entity recognition and sentiment extraction models by 18%, leading to faster client reporting workflows.
- Collaborated with a cross-functional team of 6 engineers and domain experts to deploy deep learning models on healthcare and manufacturing text datasets, achieving a 12% uplift in operational prediction accuracy.
- Built and validated causal language modeling pipelines for retail demand forecasting, reducing supply chain prediction errors by 15% across 3 major client projects.

Definity Financial, ON, Canada. Jan 2023 – May 2023
Data Scientist and Modeling (Co-op)

- Prepared image, text, and tabular data on car accidents for predictive modeling.
- Built and deployed four models for underwriting, actuary, and claims with reproducible pipelines.

Alibaba Group – Lazada E-commerce, HCMC, Vietnam. Sep 2020 – Aug 2021
Senior Manager, Category Management

- Collaborated with the Data Science team to evaluate search exposure and sales drivers (e.g., free shipping, vouchers) using text mining and A/B testing techniques.
- Conducted hypothesis-driven ad-hoc analysis to address business-critical operational queries.

HONORS & AWARDS

- CEAIS Seed Grant Award, IBM Center for Emerging Artificial Intelligence Systems. Nov 2025
- IP Protection Invention Award, College of Emergency Preparedness and Homeland Security. Oct 2025
- Travel Fund Award ($1000) from Research Foundation, SUNY Albany. Oct 2025
- Second Prize, Best Poster: *A LDP Watermark with Guaranteed Utility for LLMs* at NTIR 2025. Apr 2025
- Valedictorian, Master in Analytics – Fall 2021 Cohort, Northeastern University. Jul 2023
- Hackathon Winner – OCR and Language Model for Form Autofill, Definity Financial. Feb 2023
- Top 20 Finalist – VinUniversity Global Case Competition (VGCC), VinUniversity. Dec 2021
- Merit-based Scholarship, Foreign Trade University. 2013 & 2014
- Talent Incubation Scholarship (Top 5 students), Coca-Cola Vietnam & Thanh Nien News. Nov 2013
- Ambassador – Exchange Participant, AIESEC Indonesia. Jun – Aug 2013

PROFESSIONAL SERVICES

- **Mentor and Speaker**: South Colonie AI Superintendent's Conference Day. Sept 2026
- **Organizer committee**: Where Innovation Meets Information NTIR Conference 2025, 2026. Apr 2025/2026
- **Journal reviewer**: Journal of Combinatorial Optimization - JOCO. Jan 2025 - Present
- **Conference reviewer**: Computational Data and Social Networks - CSoNet. Oct 2024 - Present